

CTMT and the Omori Law: A Consistency Result and a Precise Limit

Why the Memory Kernel Accommodates Aftershock Decay but Cannot Validate the Framework

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Abstract

This note reports an attempted validation pillar and its exact mathematical limit, honestly. The plan was to recover a historic constant—the Omori aftershock exponent p —through the CTMT Mori–Zwanzig memory-kernel pipeline, as independent evidence for the framework. Two facts settle the attempt. First, the finite linear (Ornstein–Uhlenbeck) memory kernel of Paper IV, $K(\tau) = A_{RN}e^{A_{NN}\tau}A_{NR}$, decays *exponentially*, not as the power law $(\tau + c)^{-p}$ of Omori: a finite unresolved sector cannot produce aftershock decay. Second, the Omori law does emerge if the unresolved sector has a *continuum* of relaxation rates with spectral density $\rho(\lambda) \propto \lambda^{p-1}$ near zero, in which case the Omori exponent equals that spectral exponent exactly, $\int_0^\infty e^{-\lambda\tau} \lambda^{p-1} e^{-c\lambda} d\lambda / \Gamma(p) = (\tau + c)^{-p}$. The decisive point is that, by Bernstein’s theorem, *every* completely monotone kernel is a superposition of exponentials; the kernel formalism can therefore represent essentially any relaxation law, so reproducing Omori constrains nothing and cannot count as validation—it is a reparametrization that reads the Omori exponent as the null spectral exponent. We state this as a consistency (embedding) result, identify the single form of test that would be genuine—a cross-relation between two independently measured exponents forced by fluctuation–dissipation—and mark it as open. All statements are numerically verified.

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1 Purpose and outcome

The proposed Pillar 2 was: apply the CTMT memory-kernel machinery blind to a canonical, independently established real-world relaxation law and show it recovers the known constant.

The natural candidate is Omori’s law for aftershock decay, already flagged in the CTMT seismic note as a driven memory response.

The honest outcome is negative-as-validation and positive-as-consistency. The mathematics does not permit Omori recovery to serve as independent evidence for CTMT, for a sharp and instructive reason. We document the obstruction precisely, because a clearly delimited negative result is itself a contribution and marks the true boundary of the framework’s current empirical reach.

2 A finite unresolved sector gives exponential, not Omori, decay

Paper IV proved that eliminating the null sector from the linear dynamics yields an exact generalized Langevin equation with memory kernel

$$K(\tau) = A_{RN} e^{A_{NN}\tau} A_{NR}.$$

Proposition 2.1 (Exponential relaxation). *If A_{NN} is a finite matrix with $\text{spec}(A_{NN}) \subset \{\text{Re} < 0\}$, then $\|K(\tau)\|$ decays exponentially: $\|K(\tau)\| \leq M e^{-\gamma\tau}$ with $\gamma = -\max \text{Re spec}(A_{NN}) > 0$. In particular K cannot equal the Omori power law $(\tau + c)^{-p}$, whose ratio over successive octaves tends to a positive constant while that of an exponential tends to zero.*

Proof. $e^{A_{NN}\tau}$ is a finite sum of terms $\tau^j e^{\mu_i \tau}$ with $\text{Re } \mu_i < 0$; each is bounded by $M e^{-\gamma\tau}$. A power law is not completely captured by any finite such sum. (Verified numerically: the finite-OU kernel decays over an octave with ratio $\rightarrow 0$, unlike a power law.) $\square$

So a system with finitely many unresolved modes cannot reproduce aftershock decay. Omori is not a generic output of the linear CTMT kernel.

3 Omori emerges from a continuum null spectrum; p is its exponent

Power-law relaxation requires a broad, scale-free bath of relaxation rates.

Theorem 3.1 (Spectral origin of the Omori exponent). *Model the unresolved sector as a continuum of independent relaxation modes with rate density $\rho(\lambda)$, so the scalar memory response is $K(\tau) = \int_0^\infty e^{-\lambda\tau} \rho(\lambda) d\lambda$. If $\rho(\lambda) = \lambda^{p-1} e^{-c\lambda} / \Gamma(p)$ (a Gamma density, i.e. $\rho(\lambda) \propto \lambda^{p-1}$ as $\lambda \rightarrow 0^+$ with an upper cutoff $1/c$), then*

$$K(\tau) = (\tau + c)^{-p}.$$

Hence the CTMT kernel reproduces the Omori law exactly, and the Omori exponent p equals the small-rate exponent of the null-sector relaxation density.

Proof. $\int_0^\infty e^{-\lambda\tau} \lambda^{p-1} e^{-c\lambda} d\lambda = \Gamma(p)(\tau + c)^{-p}$ is the Gamma integral. Divide by $\Gamma(p)$. (Verified numerically to relative error $< 10^{-4}$.) $\square$

Remark 3.2. This is a real and clean identity: it maps a fitted seismological constant, p , onto a structural property of the unresolved sector—the exponent of its relaxation-rate density near zero. It is the correct CTMT statement about Omori.

4 Why this cannot validate the framework

The inability to validate CTMT through Omori is not a deficiency specific to CTMT. It follows from the general representability of completely monotone relaxation kernels established by Bernstein’s theorem. Consequently, any framework based solely on distributed exponential relaxations faces the same limitation.

Theorem 4.1 (Representability defeats validation). *By Bernstein’s theorem, a function K on $(0, \infty)$ is completely monotone if and only if it is the Laplace transform of a positive measure, $K(\tau) = \int_0^\infty e^{-\lambda\tau} d\mu(\lambda)$. Every physically admissible relaxation kernel—monotone, nonnegative, decaying, including the Omori law—is of this form. Therefore the continuum-null-sector kernel of Theorem 3.1 can represent essentially any such law by a suitable choice of ρ . Reproducing one particular law imposes no constraint that the framework could fail, so it cannot count as independent confirmation of CTMT.*

Proof. Bernstein–Widder [1]. Given any completely monotone target K , the representing measure μ exists; taking $\rho = d\mu/d\lambda$ (when absolutely continuous) realises K within the model. Matching is thus generic, not selective. $\square$

Corollary 4.2 (Consistency, not confirmation). *Recovering the Omori exponent through the CTMT kernel is a reparametrization: it re-expresses the empirical p as the null spectral exponent (Theorem 3.1). It demonstrates that the formalism is consistent with, and expressive enough to host, a real relaxation law. It does not distinguish CTMT from any other distributed-relaxation description, and therefore does not validate it.*

Non-claim 4.3 (No recovered physical constant). This note does not claim that CTMT “predicts” or “recovers” the Omori exponent as independent evidence. The exponent is an input re-read as a spectral property, not an output the theory could have gotten wrong.

5 What a genuine test would require

Validation demands a *selective* prediction: a relation the framework forces between quantities that are measured independently, so that the data could contradict it.

Definition 5.1 (Cross-exponent test). In the dynamical layer, the memory kernel K (dissipation) and the residual force covariance $\mathbb{E}[F(t)F(s)^\top]$ (fluctuation) are linked by the fluctuation–dissipation relation through the same null-sector spectrum (Paper IV). If a system supplies (i) a relaxation exponent p from the decay of K and (ii) an independent fluctuation exponent from the spectrum of the observed noise/covariance, then CTMT + fluctuation–dissipation predicts a fixed relation between the two. Agreement would be a genuine, falsifiable test; disagreement would refute the linear-stochastic modelling for that system.

Remark 5.2 (Status: open, and needs real data). Establishing the cross-exponent relation as a sharp theorem, and measuring both exponents independently in a real relaxation system (aftershock sequences, dielectric or viscoelastic relaxation, or a controlled bath), is exactly the kind of “harsh reality sample” the framework has not yet been subjected to. It is the honest location of the next real test, and it is beyond what pure theory or a flexible kernel fit can settle.

6 Conclusion

The attempt to validate CTMT by recovering the Omori constant fails, cleanly and informatively. A finite unresolved sector yields exponential, not power-law, relaxation (Proposition 2.1);

the Omori law appears only with a scale-free continuum null spectrum, whose small-rate exponent equals the Omori exponent (Theorem 3.1); and because every admissible relaxation kernel is a superposition of exponentials (Theorem 4.1), hosting Omori is generic and therefore not confirmatory (Corollary 4.2). The defensible residue is a consistency statement—CTMT’s kernel formalism is expressive enough to contain a real relaxation law and re-reads its exponent as a spectral property—together with a precise identification of the only genuine test: a fluctuation–dissipation cross-relation between two independently measured exponents, which requires real data. This is the boundary, stated exactly: the framework is consistent with the historic law, and honest about not being confirmed by it.

References

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